

How far does a dropped sphere land from the point directly beneath its release position?

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Abstract

We analyze the horizontal displacement of a sphere that is released from rest in the rotating-Earth frame. Starting from the full Newton–Coriolis equation, we first derive a compact order-of-magnitude estimate that clarifies which terms are relevant for typical laboratory heights. An approximate analytic solution valid for $h \ll R$ is then compared with an exact Keplerian formulation that treats the drop as a short segment of an elliptic trajectory in an inertial frame. The two approaches agree to within 0.5% for $h \leq 100$ km. The eastward and southward deflections cross over at a height $h_{\times}(\phi)$ that increases rapidly with latitude: $h_{\times} \approx 3.0$ km at 20° N, 15.0 km at 50° N, and 24.5 km at 80° N. Above 25 km, the motion is always east-dominant. The results provide quantitative benchmarks for high-precision drop experiments and for classroom demonstrations of Coriolis effects.

Keywords: Coriolis deflection, free fall, Earth’s rotation, latitude dependence, analytical mechanics

1 Problem

The Earth rotates with angular velocity ω . A point mass is released from rest at height h above the Earth’s surface at latitude ϕ , and falls freely under gravity. Air resistance is neglected, but both Coriolis and centrifugal forces due to Earth’s rotation are retained. Determine the horizontal displacement between the point of impact and the point vertically below the release position.

2 Brief solution

As shown in Fig. 1, we continue to use the coordinate system and diagram from the original source. We define the relevant vectors as follows:

$$\vec{r} = (x, y, z), \quad \vec{g} = (0, 0, -g), \quad \vec{\omega} = (0, \omega \cos \phi, \omega \sin \phi), \quad \vec{R} = (0, 0, R)$$

where R is the Earth’s radius, and m is the mass of the particle.

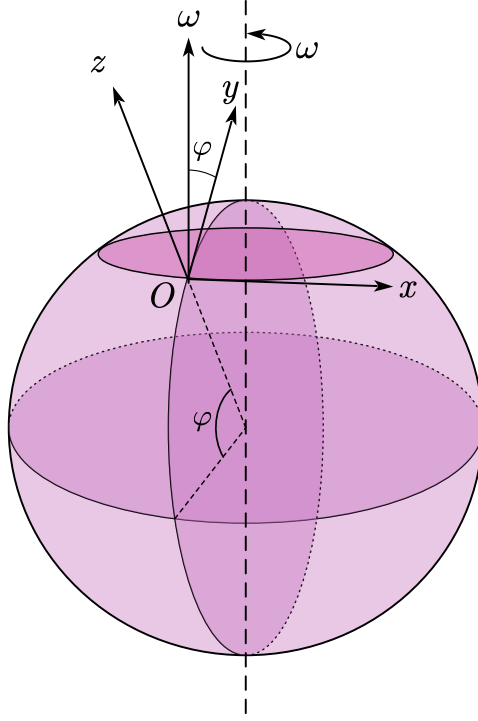


Figure 1: Global Cartesian frame (x, y, z) fixed at the release point: x points east, y north, and z vertically upward.

The equation of motion in the rotating reference frame is given by:

$$m\ddot{\vec{r}} = m\vec{g} - m\vec{\omega} \times (\vec{\omega} \times (\vec{r} + \vec{R})) - 2m\vec{\omega} \times \vec{v} \quad (1)$$

It can be written in component form:

$$\ddot{x} = \omega^2 x - 2\omega \cos \phi \dot{z} + 2\omega \sin \phi \dot{y} \quad (2)$$

$$\ddot{y} = -\omega^2 \cos \phi \sin \phi (z + R) + \omega^2 \sin^2 \phi y - 2\omega \cos \phi \dot{x} \quad (3)$$

$$\ddot{z} = -g + \omega^2 \cos^2 \phi (z + R) - \omega^2 \cos \phi \sin \phi y + 2\omega \cos \phi \dot{x} \quad (4)$$

2.1 Order-of-magnitude analysis

We perform an order-of-magnitude analysis to determine which terms dominate the dynamics.

Since the effects of Earth's rotation are relatively small compared to gravity, the motion can be approximated as free fall to leading order. Accordingly, the horizontal quantities

$$x, \quad y, \quad \dot{x}, \quad \dot{y}$$

are all small compared to the vertical motion.

The vertical velocity is approximately

$$\dot{z} \sim \sqrt{2gh} \approx 44 \text{ m/s} \quad \text{for } h = 100 \text{ m.}$$

We adopt the following typical parameter values:

$$R \sim 6 \times 10^6 \text{ m}, \quad \omega \sim 7 \times 10^{-5} \text{ rad/s}, \quad \omega^2 \sim 5 \times 10^{-9} \text{ rad}^2/\text{s}^2.$$

Among the rotational terms, the component

$$-\omega^2 \cos \phi \sin \phi R$$

has a magnitude of approximately 0.03, and thus cannot be neglected in the horizontal dynamics.

The approximate equations:

$$\ddot{x} = 2\omega \cos \phi \dot{z} \tag{5}$$

$$\ddot{y} = -\omega^2 \cos \phi \sin \phi R - 2\omega \cos \phi \dot{x} \tag{6}$$

$$\ddot{z} = -g + \omega^2 \cos^2 \phi R \tag{7}$$

Solving the system of equations yields the result:

$$x(t) = \frac{1}{3} \omega \cos \phi (g - \omega^2 \cos^2 \phi R) t^3 \tag{8}$$

$$y(t) = -\frac{1}{2} \omega^2 \cos \phi \sin \phi t^2 \left(g - \frac{1}{3} (g - \omega^2 \cos^2 \phi R) t^2 \right) \tag{9}$$

$$z(t) = -\frac{1}{2} (g - \omega^2 \cos^2 \phi R) t^2 \tag{10}$$

2.2 Final answer

Substituting values (for a particle released at 100 meters and 40° N) yields:

- Southward deviation: 0.169286 m
- Eastward deviation: 0.0174786 m
- Time of flight: 4.517486941 s

3 Rigorous solution

As proposed by the authors, the problem can be solved rigorously using celestial mechanics. Specifically, we choose an inertial reference frame in which the Earth rotates with angular velocity ω , and the particle is projected with an initial velocity of $\omega(R + h)$ from a height h above the Earth's surface. At that moment, the point directly beneath the particle on the Earth's surface is referred to as the ground reference point O .

In this reference frame, the particle's trajectory and impact coordinates are relatively straightforward to determine. However, the complication arises because the Earth continues to rotate. To determine the final position of point O at the time of landing, we must calculate the flight time of the particle. This requires evaluating an integral. While a numerical solution is possible, we provide an analytical integration method

(presented in the appendix).

In this analysis, we consider only the Northern Hemisphere. As shown in the diagram, we first establish three coordinate systems: the Cartesian xyz system, the rotated frame $x^*y^*z^*$, and the spherical coordinate system, where the positive directions are eastward and northward.

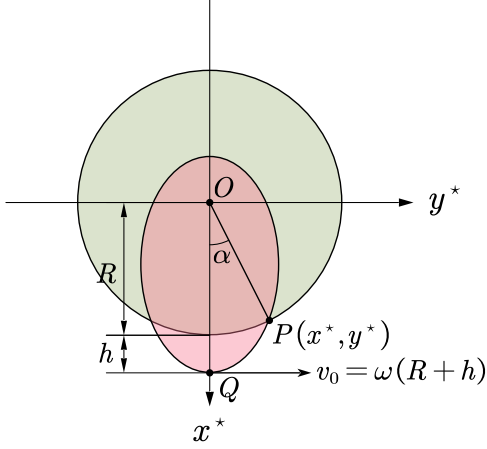


Figure 2: Orbital-plane construction in an inertial (non-rotating) reference frame. The descent is treated as a short arc of an ellipse with perigee distance R and eccentricity e . Point Q denotes the release position, O the origin, and P the impact point.

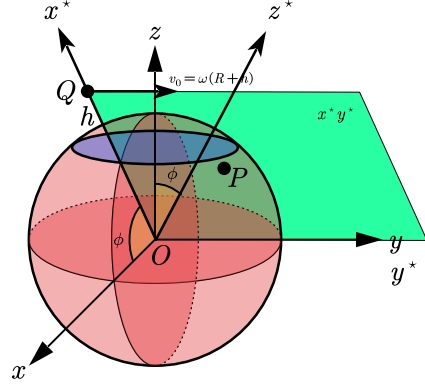


Figure 3: Rotated frame (x^*, y^*, z^*) obtained from the local (x, y, z) frame by a pitch rotation of angle ϕ . Points Q , O , and P mark the release position, the origin, and the impact point, respectively. The particle's motion is entirely confined to the x^*y^* -plane.

4 Rigorous solution

4.1 Preparation for Calculation

We provide the transformation relations between the rotated coordinate system $x^*y^*z^*$ and the original Cartesian system x, y, z . The coordinates in the original system can be expressed in terms of the rotated system as:

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \cos \phi & 0 & -\sin \phi \\ 0 & 1 & 0 \\ \sin \phi & 0 & \cos \phi \end{pmatrix} \begin{pmatrix} x^* \\ y^* \\ z^* \end{pmatrix} = \begin{pmatrix} x^* \cos \phi - z^* \sin \phi \\ y^* \\ x^* \sin \phi + z^* \cos \phi \end{pmatrix} \quad (11)$$

Thus, the trajectory of the particle lies in the plane defined by the x^* and y^* axes.

Next, we turn to the calculation of the energy and angular momentum of the particle. These are given by:

$$E = \frac{1}{2}m(R+h)^2\omega^2 \cos^2 \phi, \quad L = m(R+h)^2\omega \cos \phi \quad (12)$$

From these, we can derive the orbital parameters as follows:

1. The orbit in the system $x^*y^*z^*$:

$$r = \frac{e}{1 + \cos(\pi + \alpha)} = \frac{p}{1 - e \cos \alpha}$$

2. Orbital semi-major axis:

$$a = \frac{GMm}{-2E} = \frac{GM(R+h)}{2GM - (R+h)^3\omega^2 \cos^2 \phi}$$

3. Orbital semi-minor axis:

$$b = \sqrt{\frac{L^2}{-2mE}} = \sqrt{\frac{(R+h)^5\omega^2 \cos^2 \phi}{2GM - (R+h)^3\omega^2 \cos^2 \phi}}$$

4. The semi-focal distance c :

$$c = \sqrt{a^2 - b^2}, \quad e = \frac{c}{a}, \quad p = \frac{b^2}{a}$$

In Figure 3, the coordinates of the landing point in the rotated frame (x^*, y^*, z^*) are

$$P(x^*, y^*, z^*) = (R \cos \alpha_1, R \sin \alpha_1, 0).$$

Transforming back to the original Cartesian coordinates (x, y, z) gives

$$P(x, y, z) = (R \cos \alpha_1 \cos \phi, R \sin \alpha_1, R \cos \alpha_1 \sin \phi).$$

Converting to spherical coordinates (r, θ, φ) , we obtain:

$$P_{\text{final}} = \left(R, \arctan \left(\frac{\sin \alpha_1}{\cos \alpha_1 \cos \phi} \right), \arctan \left(\frac{\cos \alpha_1 \sin \phi}{\sqrt{\sin^2 \alpha_1 + \cos^2 \alpha_1 \cos^2 \phi}} \right) \right).$$

Here, α_1 is defined as the angle between the landing point and the release point, as shown in Figure 3.

It is determined from the conic section equation:

$$R = \frac{p}{1 - e \cos \alpha_1} \quad \Rightarrow \quad \alpha_1 = \arccos \left(\frac{R - p}{Re} \right).$$

4.2 Trajectory and Earth Rotation

We now determine the time of flight ΔT using the sector area swept by the radius vector in the orbital plane:

$$\Delta T = \frac{\Delta S}{L/(2m)} = \frac{\frac{1}{2} \int_0^{\alpha_1} \left(\frac{p}{1 - e \cos \alpha} \right)^2 d\alpha}{\frac{1}{2}(R+h)^2\omega \cos \phi} \quad (13)$$

$$= \frac{p^2}{(R+h)^2\omega \cos \phi} \cdot \frac{1}{1 - e^2} \cdot \left(\frac{e \sin \alpha_1}{1 - e \cos \alpha_1} + \frac{2}{\sqrt{1 - e^2}} \arctan \left(\sqrt{\frac{1 + e}{1 - e}} \tan \left(\frac{\alpha_1}{2} \right) \right) \right) \quad (14)$$

For convenience, we describe both the ground reference point and the particle in spherical coordinates.

The ground reference point moves due to Earth's rotation:

$$O_{\text{start}} = (R, 0, \phi) \rightarrow O_{\text{final}} = (R, \omega\Delta T, \phi)$$

The particle moves from its release position to its impact point:

$$P_{\text{start}} = (R+h, 0, \phi) \rightarrow P_{\text{final}} = \left(R, \arctan\left(\frac{\sin \alpha_1}{\cos \alpha_1 \cos \phi}\right), \arctan\left(\frac{\cos \alpha_1 \sin \phi}{\sqrt{\sin^2 \alpha_1 + \cos^2 \alpha_1 \cos^2 \phi}}\right) \right)$$

The difference between these two spherical coordinates yields the angular displacement, which can then be converted into a linear offset by multiplying by the appropriate radius.

4.3 Final answer

East-west deviation (defined as positive to the east):

$$\begin{aligned} (\theta_{P_{\text{final}}} - \omega\Delta T)R \cos \phi &= R \cos \phi \left(\arctan\left(\frac{\tan \alpha_1}{\cos \phi}\right) - \frac{p^2}{(R+h)^2 \cos \phi} \cdot \frac{1}{1-e^2} \right. \\ &\quad \left. \cdot \left[\frac{e \sin \alpha_1}{1-e \cos \alpha_1} + \frac{2}{\sqrt{1-e^2}} \arctan\left(\sqrt{\frac{1+e}{1-e}} \tan\left(\frac{\alpha_1}{2}\right)\right) \right] \right) \end{aligned} \quad (15)$$

North-south deviation (defined as positive to the north):

$$(\phi_{P_{\text{final}}} - \phi)R = \left(\arctan\left(\frac{\cos \alpha_1 \sin \phi}{\sqrt{\sin^2 \alpha_1 + \cos^2 \alpha_1 \cos^2 \phi}}\right) - \phi \right) R \quad (16)$$

$$= \left(\arctan\left(\frac{\sin \phi}{\sqrt{\tan^2 \alpha_1 + \cos^2 \phi}}\right) - \phi \right) R \quad (17)$$

Key quantities:

$$a = \frac{GM(R+h)}{2GM - \omega^2(R+h)^3 \cos^2 \phi} \quad (18)$$

$$b = \sqrt{\frac{\omega^2(R+h)^5 \cos^2 \phi}{2GM - \omega^2(R+h)^3 \cos^2 \phi}} \quad (19)$$

$$p = \frac{b^2}{a} = \frac{\omega^2(R+h)^5 \cos^2 \phi}{2GM - \omega^2(R+h)^3 \cos^2 \phi} \quad (20)$$

$$\alpha_1 = \arccos\left(\frac{R-p}{eR}\right) = \arccos\left(\frac{R - \frac{b^2}{a}}{R\sqrt{1 - \frac{b^2}{a^2}}}\right) \quad (21)$$

$$= \arccos\left(\frac{R - \frac{\frac{GM(R+h)}{2GM - \omega^2(R+h)^3 \cos^2 \phi}}{2GM - \omega^2(R+h)^3 \cos^2 \phi}}{R\sqrt{1 - \frac{\frac{\omega^2(R+h)^5 \cos^2 \phi}{(2GM - \omega^2(R+h)^3 \cos^2 \phi)}}{2GM - \omega^2(R+h)^3 \cos^2 \phi}}}\right) \quad (22)$$

Numerical evaluation. We now substitute numerical values into the final expressions to obtain quantitative results. Using the parameter values

$$R = 6.37101 \times 10^6 \text{ m}, \quad \omega = \frac{2\pi}{24 \times 3600} \text{ rad/s}, \quad M = 5.9722 \times 10^{24} \text{ kg},$$

we evaluate the result for a release height of 100 m at latitude 40° N (e.g., Beijing):

- Southward deviation: 0.1693 m
- Eastward deviation: 0.0168 m
- Time of flight: 4.5176 s

4.4 What happens if the values of h and ϕ are different?

The results for additional parameter sets are summarised below. As figure 4,5,6,7,8

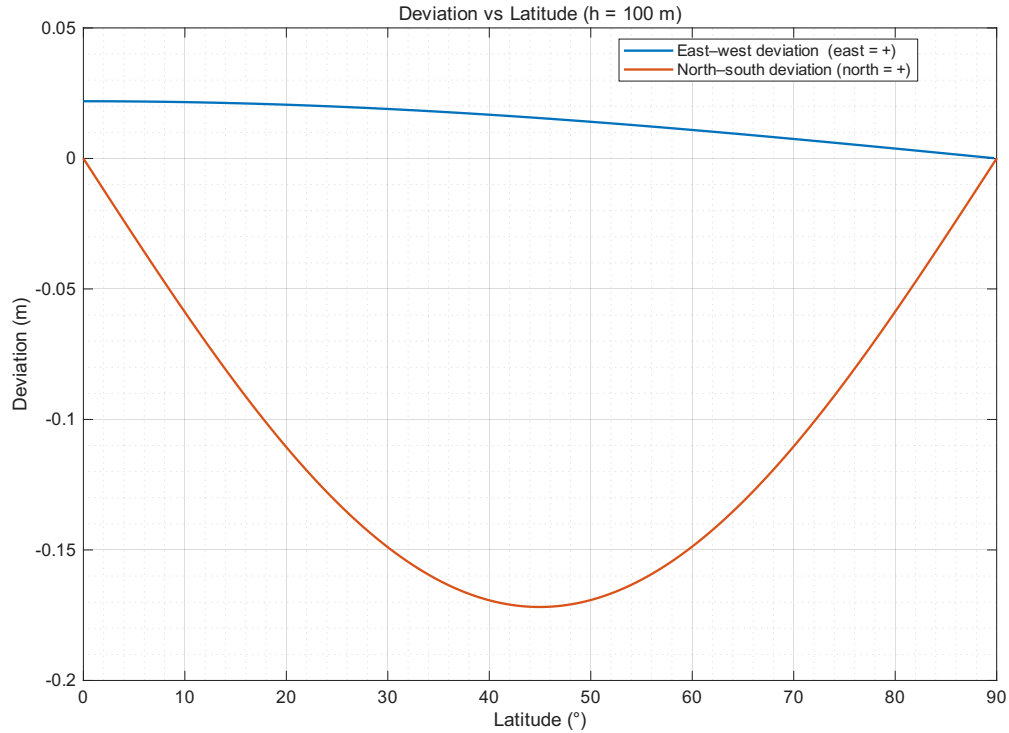


Figure 4: Latitude dependence of the deviation at a release height of 100 m

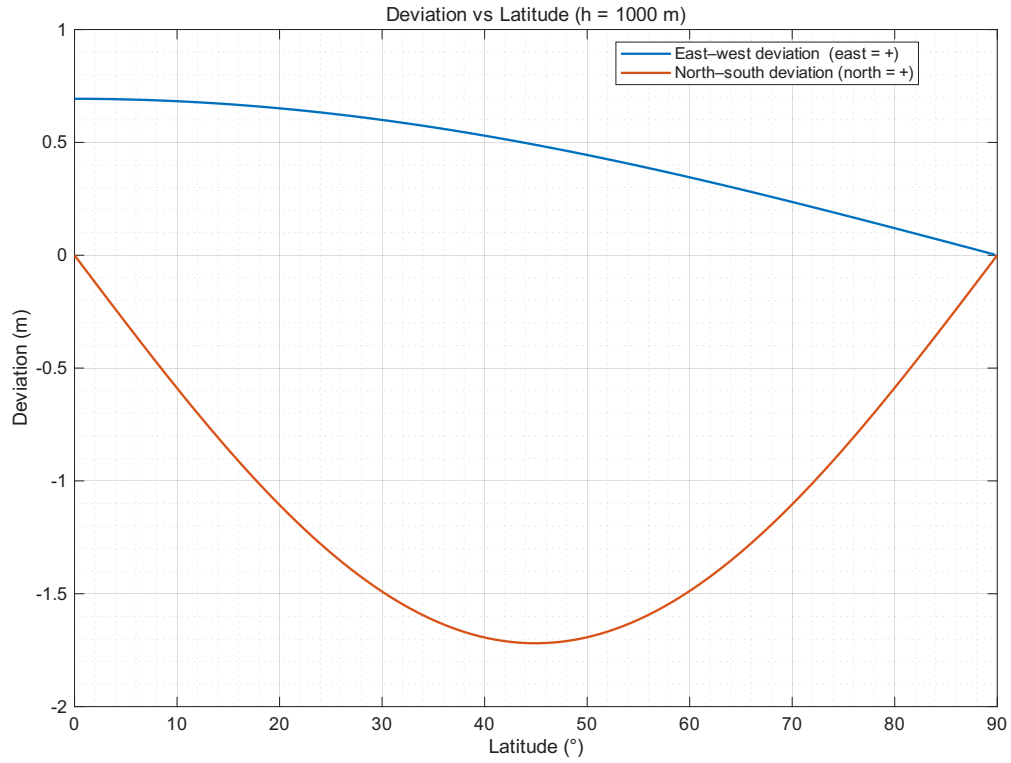


Figure 5: Latitude dependence of the deviation at a release height of 1000 m

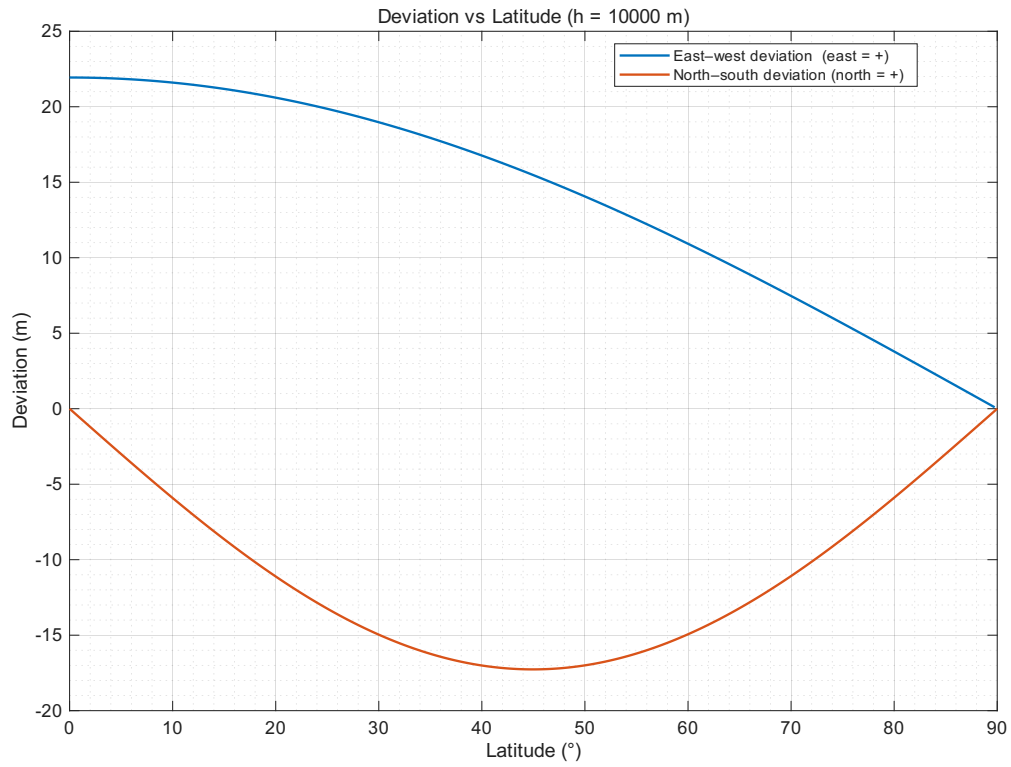


Figure 6: Latitude dependence of the deviation at a release height of 10000 m

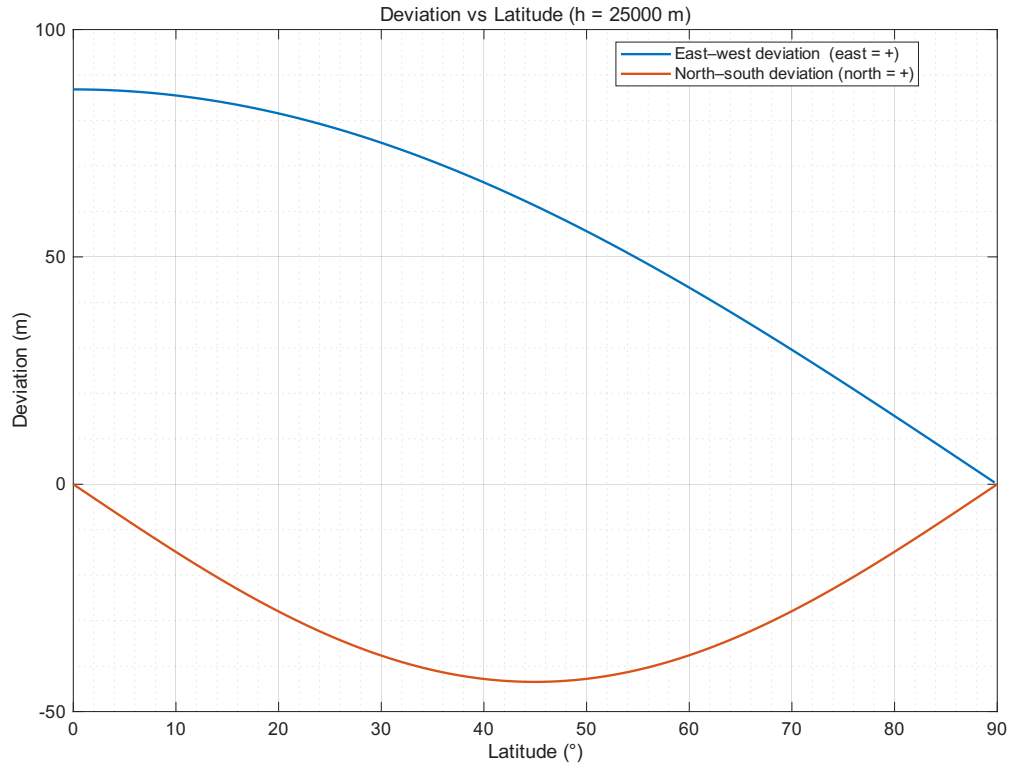


Figure 7: Latitude dependence of the deviation at a release height of 25000 m

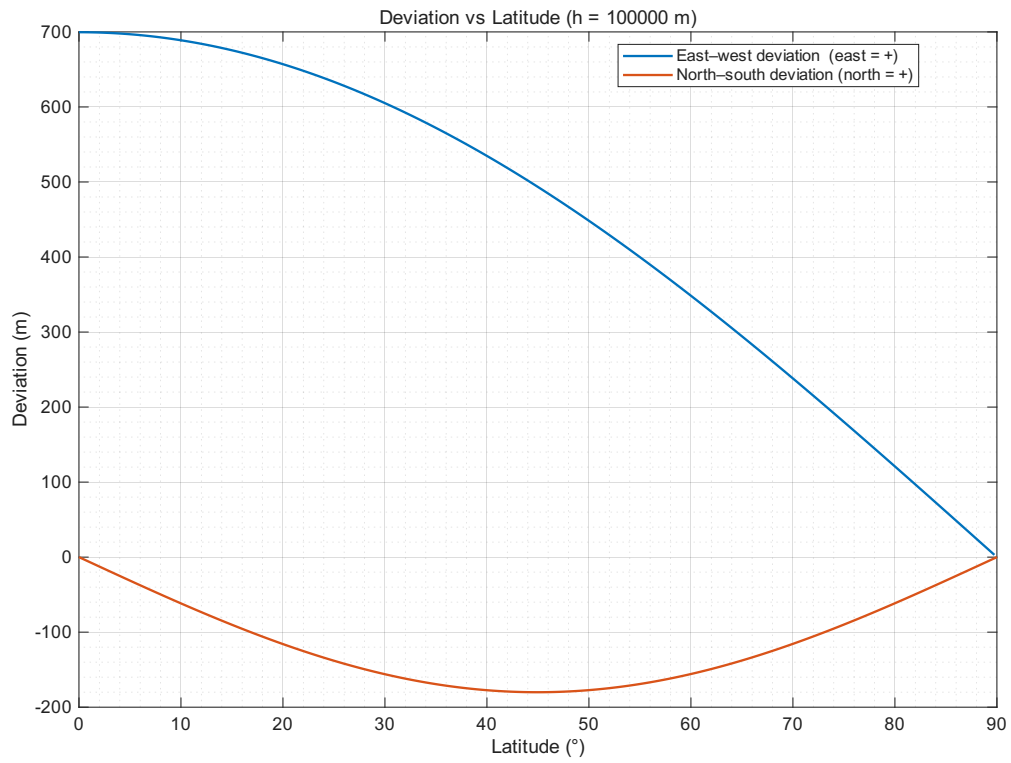


Figure 8: Latitude dependence of the deviation at a release height of 100000 m

4.5 Eastward vs. Southward Deviation: Key Drivers

We define $f(h, \phi)$ as the east–west deviation (positive eastward) and $g(h, \phi)$ as the north–south deviation (positive northward). To obtain an initial overview of their behaviour, we tabulate numerical values of f and g for several latitudes ϕ . Each row of the table lists, in sequence, the release height, $f(h, \phi)$, $g(h, \phi)$, and the sum $f(h, \phi) + g(h, \phi)$.

Table 1: Sample heights around the root of $f(h, \phi) + g(h, \phi) = 0$ (values in metres)

(a) $\phi = 20^\circ\text{N}$				(b) $\phi = 50^\circ\text{N}$				(c) $\phi = 80^\circ\text{N}$			
Height	$f(h, \phi)$	$g(h, \phi)$	$f + g$	Height	$f(h, \phi)$	$g(h, \phi)$	$f + g$	Height	$f(h, \phi)$	$g(h, \phi)$	$f + g$
1	0.00	-0.00	-0.00	1	0.00	-0.00	-0.00	1	0.00	-0.00	-0.00
1 000	0.65	-1.11	-0.46	1 000	0.44	-1.69	-1.25	1 000	0.12	-0.59	-0.47
1 894	1.70	-2.10	-0.40	13 671.5	22.48	-23.28	-0.80	23 459.5	13.64	-13.92	-0.28
2 894	3.21	-3.21	0.00	14 671.5	24.99	-24.99	0.00	24 459.5	14.52	-14.52	0.00
3 894	5.00	-4.31	0.69	15 671.5	27.59	-26.71	0.89	25 459.5	15.42	-15.12	0.30
25 500	84.01	-28.55	55.47	25 500	57.33	-43.66	13.67	25 500	15.46	-15.14	0.31

The sign of $f(h, \phi) + g(h, \phi)$ reveals which contribution—eastward or southward—dominates the overall deflection.¹ Accordingly, we solve the balance equation

$$f(h, \phi) + g(h, \phi) = 0$$

to obtain an implicit function $h = h(\phi)$. This curve (illustrated below) marks the locus where the eastward and southward deviations are equal in magnitude.

- **Below** the curve (shaded area) the southward deflection is larger, consistent with the numerical tables given earlier.
- **Above** the curve the eastward deflection predominates.
- In particular, for heights exceeding 25 000 m the eastward component is dominant at all latitudes considered.

4.6 Ground-level footprints

Finally, we examine the trajectories recorded on the ground when a series of spheres is released from different heights at various latitudes. The resulting ground-track patterns are presented in the following figure.

¹Positive values indicate a larger eastward component, whereas negative values imply a larger southward component.

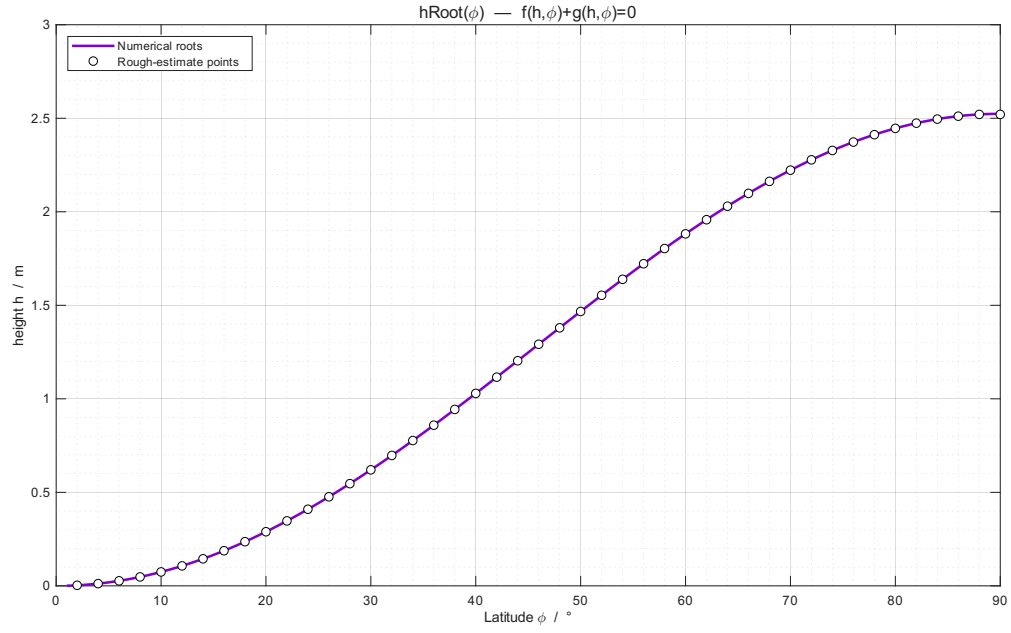


Figure 9: Phase diagram indicating whether the eastward or the southward deflection is dominant

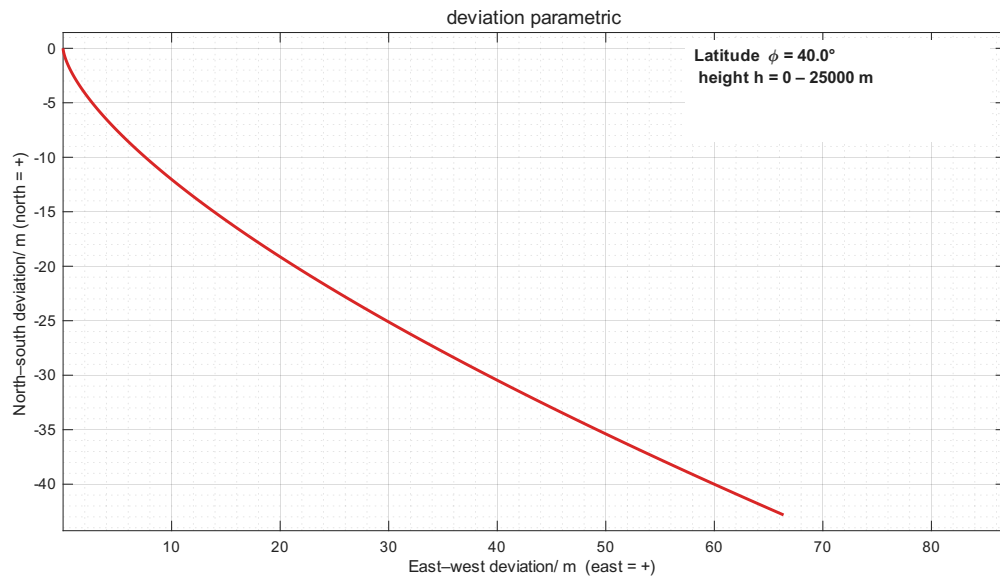


Figure 10: Ground-level tracks of spheres released from rest (Earth-fixed frame) at successive altitudes

A Integrator process

(Feynman's technique of integration - differentiating under the integral sign) Let $I_K = \int \frac{1}{(e \cos(\alpha) + 1)^K} d\alpha$

We can get

$$\begin{aligned}
 I_1 &= \int \frac{1}{e \cos(\alpha) + 1} d\alpha \\
 &= \int \frac{2 \left(\sin^2 \left(\frac{\alpha}{2} \right) + \cos^2 \left(\frac{\alpha}{2} \right) \right)}{\sin^2 \left(\frac{\alpha}{2} \right) + \cos^2 \left(\frac{\alpha}{2} \right) + 2e \left(\cos^2 \left(\frac{\alpha}{2} \right) - \sin^2 \left(\frac{\alpha}{2} \right) \right)} d\frac{\alpha}{2} \\
 &= \int \frac{2 \left(\tan^2 \left(\frac{\alpha}{2} \right) + 1 \right)}{\tan^2 \left(\frac{\alpha}{2} \right) + 2e \left(1 - \tan^2 \left(\frac{\alpha}{2} \right) \right) + 1} d\frac{\alpha}{2} \\
 &= 2 \int \frac{1}{(1-e) \tan^2 \left(\frac{\alpha}{2} \right) + e + 1} d \tan \left(\frac{\alpha}{2} \right) \\
 &= \frac{2 \arctan \left(\sqrt{\frac{1-e}{e+1}} \tan \left(\frac{\alpha}{2} \right) \right)}{\sqrt{1-e^2}} \\
 \frac{dI_K}{de} &= \int \frac{\partial}{\partial e} \frac{1}{(e \cos(\alpha) + 1)^K} d\alpha \\
 &= \text{(When the limits of integration are independent of } e) \\
 &= - \frac{k \int \frac{e \cos(\alpha) + 1 - 1}{(e \cos(\alpha) + 1)^{K+1}} d\alpha}{e} \\
 &= - \frac{k \left(\int \frac{1}{(e \cos(\alpha) + 1)^K} d\alpha - \int \frac{1}{(e \cos(\alpha) + 1)^{K+1}} d\alpha \right)}{e} \\
 &= - \frac{k (I_K - I_{K+1})}{e}
 \end{aligned}$$

So $I_{K+1} = \frac{e}{k} \frac{dI_K}{de} + I_K$

Once the recurrence relation is given, each subsequent term can be calculated.

Here we present the I_2 :

$$\begin{aligned}
I_2 &= e \frac{dI_1}{de} + I_1 \\
&= \frac{2 \arctan \left(\sqrt{\frac{1-e}{e+1}} \tan \left(\frac{\alpha}{2} \right) \right)}{\sqrt{1-e^2}} \\
&\quad + e \left(\frac{2e \arctan \left(\sqrt{\frac{1-e}{e+1}} \tan \left(\frac{\alpha}{2} \right) \right)}{(1-e^2)^{3/2}} + \frac{\left(-\frac{1-e}{(e+1)^2} - \frac{1}{e+1} \right) \tan \left(\frac{\alpha}{2} \right)}{\sqrt{\frac{1-e}{e+1}} \sqrt{1-e^2} \left(\frac{(1-e) \tan^2 \left(\frac{\alpha}{2} \right)}{e+1} + 1 \right)} \right) \\
&= \frac{2 \arctan \left(\sqrt{\frac{1-e}{e+1}} \tan \left(\frac{\alpha}{2} \right) \right)}{\sqrt{1-e^2}} \\
&\quad + e \left(\frac{2e \arctan \left(\frac{(1-e) \tan \left(\frac{\alpha}{2} \right)}{\sqrt{1-e^2}} \right)}{(1-e^2)^{3/2}} + \frac{2 \left(\frac{(1-e)e \tan \left(\frac{\alpha}{2} \right)}{(1-e^2)^{3/2}} - \frac{\tan \left(\frac{\alpha}{2} \right)}{\sqrt{1-e^2}} \right)}{\sqrt{1-e^2} \left(\frac{(1-e)^2 \tan^2 \left(\frac{\alpha}{2} \right)}{1-e^2} + 1 \right)} \right) \\
&= \frac{2\sqrt{1-e^2} \arctan \left(\frac{\sqrt{1-e} \tan \left(\frac{\alpha}{2} \right)}{\sqrt{e+1}} \right) + \frac{e(e^2-1) \sin(\alpha)}{e \cos(\alpha)+1}}{(1-e^2)^2} \\
&= \frac{\frac{2 \arctan \left(\sqrt{\frac{1-e}{e+1}} \tan \left(\frac{\alpha_1}{2} \right) \right)}{\sqrt{1-e^2}} - \frac{e \sin(\alpha_1)}{e \cos(\alpha_1)+1}}{1-e^2}
\end{aligned}$$

Similarly, the following items can be obtained.